

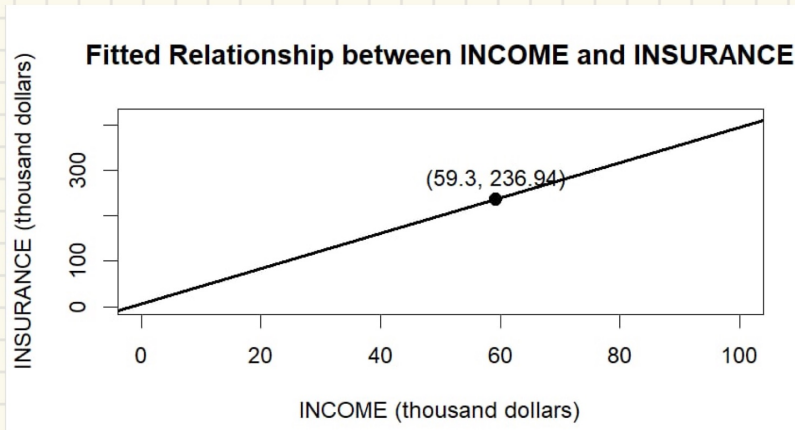
- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880 INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

a)



$$\widehat{insurance} = 6,855 + 3,880 \times 59,3 = 236,94$$

b)

$$\Delta insurance = 3,880 \times 1000 = \$3880$$

→ Increase \$3880 insurance with each additional \$1000 of household income.

$$df = 20 - 2 = 18$$

$$t_{0,975,18} = 2,101$$

$$CI = \hat{\beta}_1 \pm t_{0,975,18} \times se(\hat{\beta}_1)$$

$$= 3,880 \pm 2,101 \times 0,112 \Rightarrow [3,645; 4,115]$$

In repeated sampling, about 95% intervals constructed this way will contain the true value of parameter β_1

OR: With 95% confidence, each additional \$1000 income increases insurance by \$3645 to \$4115.

- Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

c. income = 100 (thousands \$)

$$\widehat{\text{insurance}} = 6,855 + 2,880 \times 100 = 394,855$$

$$\begin{aligned} \text{se}(\hat{y}) &= \sqrt{\text{Var}(\hat{\beta}_0) + x^2 \text{Var}(\hat{\beta}_1) + 2x \text{cov}(\hat{\beta}_0, \hat{\beta}_1)} \\ &= \sqrt{7,383^2 + 100^2 \cdot 0,112^2 + 2 \cdot 100 \cdot (-0,746)} \\ &= 5,545 \end{aligned}$$

$$t_{0,995,18} = 2,878$$

$$\begin{aligned} \text{CI} &= \hat{y} \pm t_{0,995,18} \times \text{se}(\hat{y}) \\ &= 394,855 \pm 2,878 \times 5,545 \Rightarrow [378,90; 410,81] \end{aligned}$$

d)

Hypotheses :

$$H_0 : \beta_1 = 5$$

$$H_1 : \beta_1 \neq 5$$

Test statistic :

$$t_{\text{stat}} = \frac{b_1 - 5}{\text{se}(b_1)} = \frac{3,880 - 5}{0,112} = -10$$

Critical value

$$t_{0,975,18} = 2,101$$

$$\Rightarrow |t| > 2,101 \quad (10 > 2,101) \Rightarrow \text{reject } H_0$$

$\Rightarrow$ We reject the hypothesis that the insurance increases by \$ 5000 per \$1000 income.

e)

Hypotheses :

$$H_0 : \beta_1 = 1$$

$$H_1 : \beta_1 > 1$$

$$\text{Test statistic} = t = \frac{b_1 - 1}{\text{se}(b_1)} = \frac{3,880 - 1}{0,112} = 25,71$$

$$\text{Critical value} : t_{0,99,18} = 2,552$$

$$\Rightarrow t > 2,552 \quad (25,71 > 2,552) \Rightarrow \text{reject } H_0$$

$\Rightarrow$ We reject the hypothesis that income increases the amount of life insurance held increase by the same amount, and conclude that the income increases the amount of life insurance held higher than the amount itself.

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test *p*-value. What do you conclude?
- Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

3.23

a)

$$PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$$

$$\Rightarrow \frac{\Delta PRICE}{\Delta SQFT} = 2\alpha_2 SQFT$$

$$PRICE = 13 \text{ (thousand \$)}$$

$$SQFT = 20 \text{ (hundred sqft)}$$

$$\frac{\Delta PRICE}{\Delta SQFT} = 2 \cdot 20 \cdot \alpha_2 = 40\alpha_2$$

$$\text{Hypotheses: } H_0: 40\alpha_2 \leq 13 \Leftrightarrow H_0: \alpha_2 \leq 0,325$$

$$H_1: 40\alpha_2 > 13 \Leftrightarrow H_1: \alpha_2 > 0,325$$

3.23(a)

The test statistic is $t_{\text{stat}} = (a2_{\text{hat}} - 0.325) / se_{a2} = -26.72875$

The Critical value is $t_{\text{crit}} = 1.647919$

The rejection region is $t > 1.647919$

The *p*-value is $p_{\text{value}} = 1 > 0.05$

Fail to reject H_0 at 5% level of significance

There is no statistical evidence that the marginal effect of increasing house size by 100 sqft at 2000 sqft exceeds \$13000

b)

$$\frac{\Delta PRICE}{\Delta SQFT} = 2 \cdot 40 \alpha_2 = 80$$

$$\text{Hypotheses: } H_0: 80\alpha_2 \leq 13 \Leftrightarrow H_0: \alpha_2 \leq 0,1625$$

$$H_1: 80\alpha_2 > 13 \Leftrightarrow H_1: \alpha_2 > 0,1625$$

3.23(b)

The test statistic is $t_{\text{stat}} = (a2_{\text{hat}} - 0.1625) / se_{a2} = 4.189467$

The Critical value is $t_{\text{crit}} = 1.647919$

The rejection region is $t > 1.647919$

The *p*-value is $p_{\text{value}} = 1.65395e-05 < 0.05$

Reject H_0 at 5% level of significance

There is statistical evidence that the marginal effect will be greater than \$13000

test *p*-value. What do you conclude?

- Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

c)

$$E(PRICE | SQFT) = \alpha_1 + \alpha_2 (20)^2 = 95,56585 + 400 \cdot 0,184519 = 167,37$$

3.23(c)

The estimated expected price at 2000 sqft is $\hat{y}=167.3735$

The 95% CI is [154.7555, 179.9914]

In repeated sampling, about 95% intervals constructed this way will cover the true value of average price of a 2000 sqft house

3.23(d)

Number of houses

Sample average p.

The sample mean is within the 95% CI, so it is compatible with part (c).

3.31(a)

SAL1: mean=6718.712, min=1762, max=32820, sd=6915.083

APR1: mean=0.7825, min=0.61, max=0.92, sd=0.09803711

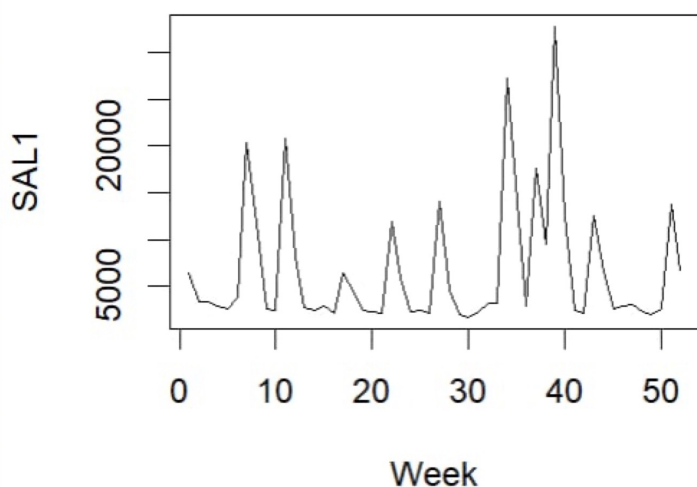
3.31

3.31(a)

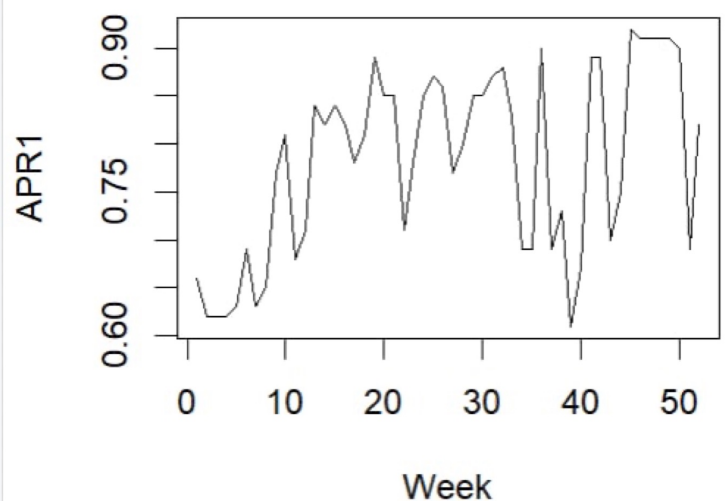
SAL1: mean=6718.712, min=1762, max=32820, sd=6915.083

APR1: mean=0.7825, min=0.61, max=0.92, sd=0.09803711

Sales (SAL1) over weeks



Price (APR1) over weeks

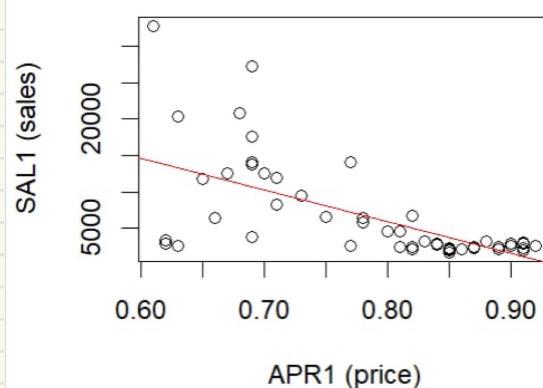


3.31(b)

Correlation between price and sales = -0.6159284

The relationship is expected to be negative (law of demand).

SAL1 vs APR1



3.31(c)

The estimated effect of a 1-cent increase in price on sales is $b_2\text{-hat} = -434.4473$

The Critical value is $t_{\text{crit}} = 2.008559$

The 95% confidence interval is $[-592.2897, -276.6049]$

3.31(d)

Point estimate for expected sales when price = 70 cents is $y\text{-hat} = 10302.9$

The Critical value is $t_{\text{crit}} = 1.675905$

The standard error of expected value is $se_{y\text{-hat}} = 1001.231$

The 90% confidence interval is $[8624.934, 11980.87]$

3.31(e)

The estimated elasticity at the sample means is $elas\text{-hat} = -5.059825$

The 95% confidence interval is $[-6.898149, -3.221501]$

Sales are fairly elastic with respect to price.

This makes economic sense because demand for a specific brand usually falls when its price rises.

3.31(f)

H_0 : elasticity = -3

H_1 : elasticity $\neq$ -3

The test statistic is $t_{\text{stat}} = -2.250572$

The critical value is ± 1.675905

The rejection region is $t < -1.675905$ or $t > 1.675905$

The p-value is 0.02884204

Reject H_0 at the 10% significance level.